

Can Graph Neural Networks Use Breast MRI Vessel Networks to Predict Treatment Response?

SPENCER VENANCIO

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Can vessel networks help personalize treatment?

THE TREATMENT

Neoadjuvant therapy is given before surgery to shrink the tumor.

THE OUTCOME

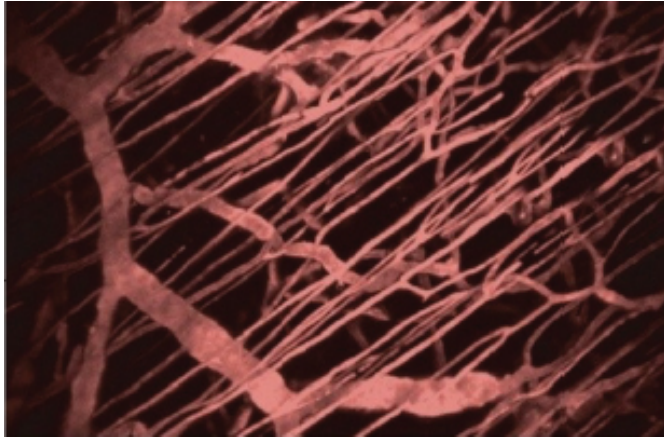
About 1 in 3 exams came from patients who achieved a **pathologic complete response** (pCR): no invasive cancer remained in the breast or nearby lymph nodes at surgery. Achieving pCR is associated with better long-term outcomes.

THE QUESTION

A **graph neural network**, or **GNN**, learns from both measurements at vessel locations and the connections between them. Can a GNN use connections within breast vessel networks to improve treatment-response prediction?

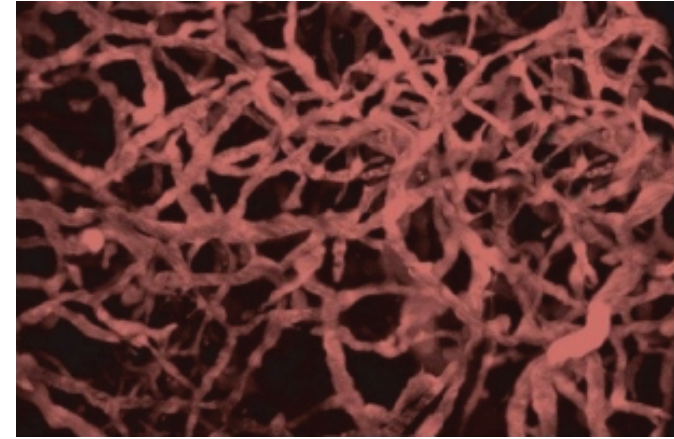
Earlier prediction could guide less intensive therapy for patients likely to achieve pCR and alternative therapy for those unlikely to do so.

Tumors build a messier blood supply



Normal vessels: organized, efficient.

Blood flows through a predictable, well-structured network.



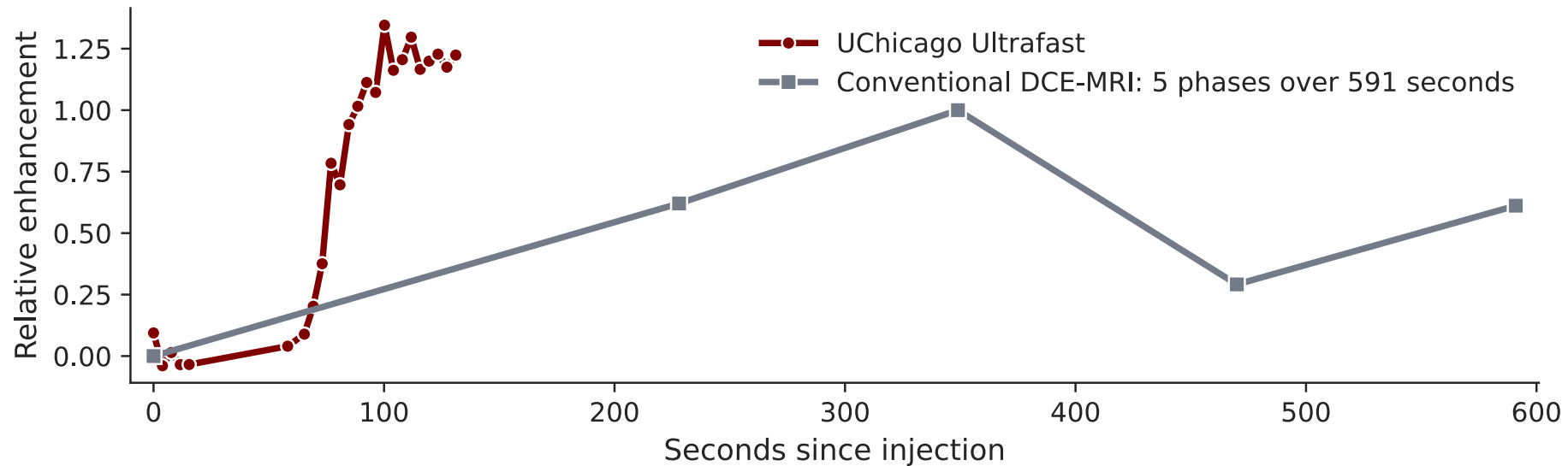
Tumor vessels: dense, tortuous, leaky.

Tumors stimulate abnormal vessel growth as they grow. Structure may reflect how aggressive a tumor is.

Because tumor vessels differ in both structure and blood flow, those patterns may help predict treatment response.

Vessel diagrams from *Nature Medicine*, Jain RK, "Molecular regulation of vessel maturation", 2003;9(6):685–693,¹⁵ copyright (2003).

Ultrafast MRI captures images every few seconds as contrast enters tissue



Curves are from different patients and are shown only to compare temporal sampling.

179

pCR-labeled exams, 141 patients

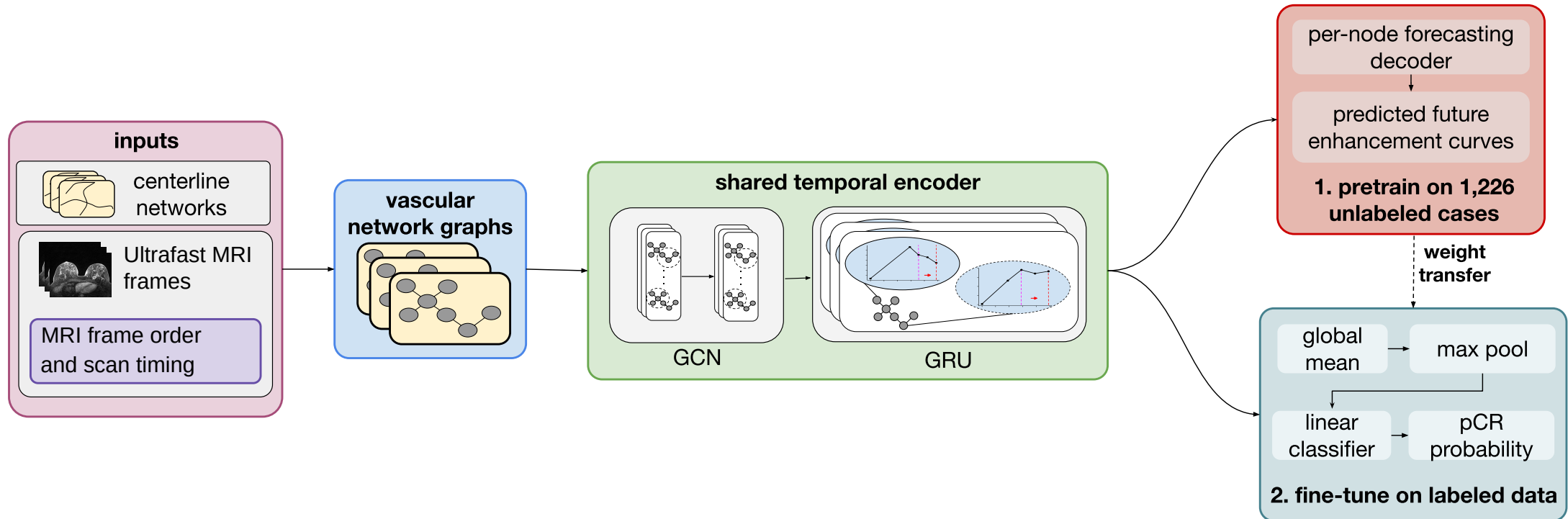
1,226

pCR-unlabeled exams, pretraining only

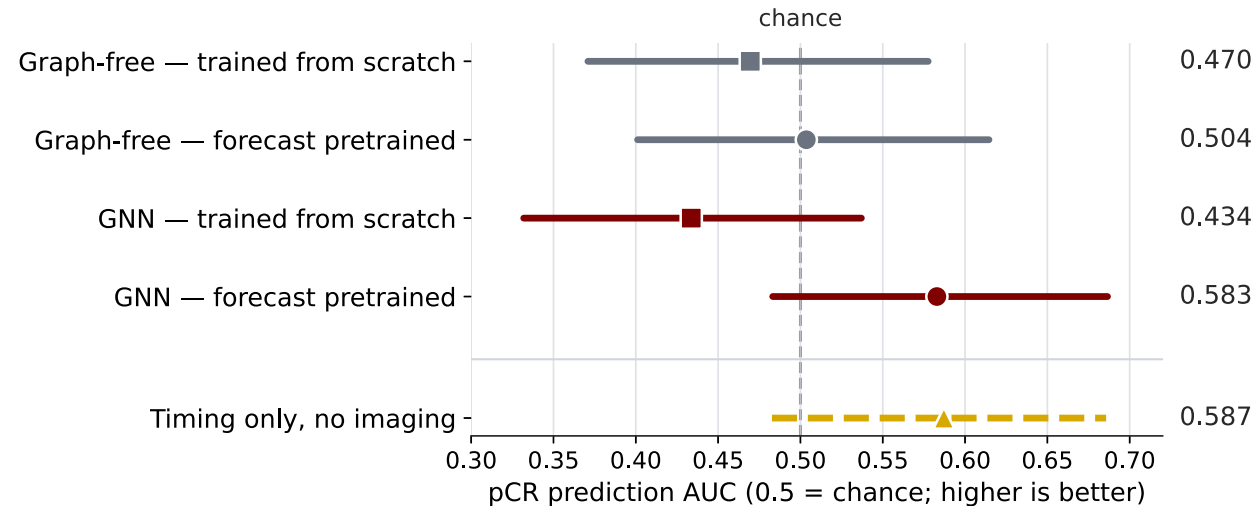
13–24

frames per ultrafast movie

The GNN first predicts later MRI frames, then predicts treatment response



Learning to predict later MRI frames improved the GNN



Pretraining helped.

Learning to predict later MRI frames improved the GNN by **0.149 AUC** (95% CI, 0.046–0.255). After pretraining, the GNN also outperformed the matched graph-free model by **0.079** (95% CI, 0.010–0.155).

But scan timing was associated with pCR too.

A model using only scan-timing information, no imaging, reached **AUC 0.587** (95% CI, 0.483–0.686), and its predictions correlated with the GNN's at **$r = 0.50$** . We don't yet know why; next we need to determine what drives that association and whether it influences the GNN's predictions.

Summary

1

Learning to predict later MRI frames **improved the GNN.**

2

With pretraining, the GNN outperformed the model that **ignored vessel connections.**

3

How the scans were timed also predicted pCR, so we need to learn whether the model is detecting **tumor biology or scan differences.**

Learning to predict contrast changes improved the GNN, but some of its signal may come from how the scans were collected.

Thank you

Code

github.com/dsi-clinic/vanguard

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